

Task 2: Bayesian Change Point Detection

Brent Oil Price Analysis

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Executive Summary

This report presents the application of Bayesian change point detection to identify structural breaks in Brent oil price returns from 1987-2022. Using PyMC for Bayesian inference, we implemented a discrete change point model that detects significant shifts in the mean of log returns. The analysis reveals one major structural break, which we associate with key geopolitical and economic events. This approach provides a statistically rigorous framework for understanding market regime changes and their relationship to major world events.

Methodology

Bayesian Change Point Model:

- Discrete uniform prior for change point location (τ)
- Normal priors for means before and after change point
- Half-Cauchy prior for standard deviation
- Normal likelihood with switching mean function

MCMC Sampling:

- Metropolis sampler for discrete parameters
- 10,000 draws with 2,000 tuning steps
- 4 parallel chains for convergence assessment
- R-hat statistics for convergence diagnostics

Data Preparation:

- Daily Brent oil prices (1987-2022)
- Log returns for statistical properties
- Event dataset for association analysis

Key Findings

Structural Break Detection:

- Single significant change point detected in the return series
- Posterior distribution provides uncertainty quantification
- 95% highest density interval for change point location

Statistical Impact:

- Quantified difference in mean returns before and after break

- Percentage change in return characteristics
- Model convergence diagnostics confirm reliable inference

Event Association:

- Major geopolitical events correlated with detected change point
- Context provided by historical oil market events
- Temporal proximity analysis for event attribution

Technical Implementation

Model Specification:

$\tau \sim \text{DiscreteUniform}(0, N-1)$
 $\mu_{\text{before}} \sim \text{Normal}(0, 0.01)$ # Mean before change point
 $\mu_{\text{after}} \sim \text{Normal}(0, 0.01)$ # Mean after change point
 $\sigma \sim \text{HalfCauchy}(0, 0.01)$ # Standard deviation
 $y \sim \text{Normal}(\mu_{\text{switch}}, \sigma)$ # Likelihood with switching mean

Computational Details:

- PyMC 4.x for probabilistic programming
- ArviZ for diagnostics and visualization
- MCMC sampling with Metropolis algorithm
- Convergence assessment via R-hat and ESS

Limitations and Extensions

Current Limitations:

- Single change point assumption (real markets have multiple breaks)
- Mean-only modeling (ignores volatility changes)
- Independence assumption (ignores autocorrelation)
- No exogenous variables in current model

Proposed Extensions:

- Multiple change point detection
- Joint mean-volatility change point models
- Regime-switching models with Markov dynamics
- Integration of event indicators and economic variables
- Real-time change point detection algorithms
- Machine learning approaches for pattern recognition

Practical Applications

Risk Management:

- Early warning system for market regime changes
- Dynamic risk model adjustments
- Stress testing scenarios based on historical breaks

Trading Strategies:

- Strategy adaptation around detected change points
- Portfolio rebalancing triggers
- Volatility forecasting improvements

Policy Analysis:

- Quantifying market impacts of policy decisions
- Event study methodology enhancement

- Economic impact assessment framework

Conclusions

The Bayesian change point analysis successfully identified a significant structural break in Brent oil price returns, demonstrating the power of probabilistic programming for financial time series analysis. The integration of statistical modeling with historical event context provides a comprehensive framework for understanding market dynamics. While the current single change point model has limitations, it establishes a foundation for more sophisticated analyses that could incorporate multiple breaks, volatility changes, and exogenous factors. This approach bridges the gap between statistical detection and economic interpretation, providing valuable insights for researchers, traders, and policymakers interested in understanding the dynamics of oil price movements and their relationship to global events.